

PROJECT 1 — Enterprise HR AI Assistant Portal

Problem Statement

Employees struggle to:

- Find HR policies
- Apply leave
- Access onboarding information
- Contact HR

Web Application

Name

HRGPT Enterprise Portal

AI Agents

Agent	Responsibility
HR FAQ Agent	HR questions
Leave Assistant Agent	Leave workflows
Onboarding Agent	Employee onboarding
Policy Search Agent	RAG document search

Tech Stack

Layer	Technology
Frontend	React + Tailwind
Backend	Node.js + Express
AI	Azure OpenAI
Authentication	Azure Entra ID
Database	MongoDB

Layer	Technology
RAG	Azure AI Search
Workflow	Power Automate

Features

- AI HR chatbot
 - Leave management
 - Employee onboarding
 - Teams integration
 - SharePoint grounding
 - AI-generated responses
 - Adaptive cards
-

Folder Structure

```
hr-ai-portal/  
├── client/  
├── server/  
├── agents/  
├── workflows/  
├── rag/  
└── auth/
```

Frontend React Starter

App.js

```
import React from "react";  
import ChatWindow from "../components/ChatWindow";
```

```
function App() {  
  return (  
    <div className="min-h-screen bg-gray-100">  
      <h1 className="text-3xl font-bold text-center p-5">  
        HRGPT Enterprise Portal  
      </h1>  
    </div>  
  );  
}
```

```

    <ChatWindow />
  </div>
);
}

```

```
export default App;
```

Chat Component

ChatWindow.js

```

import React, { useState } from "react";
import axios from "axios";

function ChatWindow() {
  const [message, setMessage] = useState("");
  const [response, setResponse] = useState("");

  const sendMessage = async () => {
    const res = await axios.post("http://localhost:5000/chat", {
      message,
    });

    setResponse(res.data.reply);
  };

  return (
    <div className="p-5">
      <textarea
        className="border p-2 w-full"
        rows="4"
        onChange={(e) => setMessage(e.target.value)}
      />

      <button
        onClick={sendMessage}
        className="bg-blue-500 text-white px-5 py-2 mt-2"
      >
        Ask HR Agent
      </button>

      <div className="mt-5 bg-white p-5 rounded shadow">

```

```

        {response}
      </div>
    </div>
  );
}

export default ChatWindow;

```

Backend API

server.js

```

const express = require("express");
const cors = require("cors");
const axios = require("axios");

const app = express();

app.use(cors());
app.use(express.json());

app.post("/chat", async (req, res) => {

  const userMessage = req.body.message;

  try {

    const response = await axios.post(

      "https://YOUR_OPENAI_ENDPOINT/openai/deployments/YOUR_MODEL/chat/completions?api-version=2024-02-15-preview",
      {
        messages: [
          {
            role: "system",
            content: "You are an enterprise HR assistant."
          },
          {
            role: "user",
            content: userMessage
          }
        ],
        temperature: 0.7
      }
    );
  } catch (error) {
    console.error("Error in chat API:", error);
    res.status(500).json({ error: "Internal server error" });
  }
});

```

```

    },
    {
      headers: {
        "api-key": process.env.AZURE_OPENAI_KEY,
        "Content-Type": "application/json"
      }
    }
  );

  res.json({
    reply: response.data.choices[0].message.content
  });

} catch (error) {
  console.log(error);

  res.status(500).json({
    error: "AI request failed"
  });
}
});

app.listen(5000, () => {
  console.log("Server running on port 5000");
});

```